

Total No. of Questions : 9]

SEAT No. :

PD-4045

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[6402]-4

S.E. (Civil)

ENGINEERING MATHEMATICS -III

(2019 Pattern) (Semester-III) (207001)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Question no. 1 is compulsory.
- 2) Attempt Q.2 or Q.3, Q.4 or Q.5, Q.6 or Q.7 and Q.8 or Q.9.
- 3) Assume Suitable data if necessary.
- 4) Neat diagrams must be drawn whenever necessary.
- 5) Figures to the right indicate full marks.
- 6) Use of electronic pocket Calculator is allowed.

Q1) a) If $f(x) = \frac{1}{\sqrt{6\pi}} e^{-x^2/6}$, $x \in (-\infty, \infty)$ then the mean μ and variance σ^2 are [1]

- i) $\mu = 1, \sigma^2 = 3$
- ii) $\mu = 0, \sigma^2 = 3$
- iii) $\mu = 0, \sigma^2 = \sqrt{3}$
- iv) $\mu = 3, \sigma^2 = 1$

b) The regression lines y on x is given by $2x - 3y + 5 = 0$. The slope b_{xy} of the regression line x on y satisfies. [2]

- i) $b_{xy} = 1$
- ii) $b_{xy} < 3/2$
- iii) $b_{xy} b_{yx} = 1$
- iv) $b_{xy} b_{yx} > 1$

P.T.O.

c) If the vector field $\vec{F} = m(x+z)\vec{i} - 2(y+z)\vec{j} - z\vec{k}$ is solenoidal then value of m is [2]

- i) -3 ii) 3
iii) 4 iv) 0

d) Unit vector along the direction of line $\frac{x}{-1} = \frac{y-2}{3} = \frac{z+1}{2}$ is [1]

- i) $\frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{3}}$ ii) $\frac{\vec{i} + 3\vec{j} + 2\vec{k}}{\sqrt{14}}$
iii) $\frac{-\vec{i} + 3\vec{j} + 2\vec{k}}{\sqrt{14}}$ iv) $\frac{-\vec{i} + 3\vec{j} + \vec{k}}{\sqrt{11}}$

e) The value of $\iiint_V \nabla \cdot \vec{F} dv$ where $\vec{F} = yz\vec{i} + xz\vec{j} + xy\vec{k}$ over the surface of sphere is [2]

- i) 3 ii) 0
iii) 4 iv) 10

f) The most general solution of $\frac{\partial u}{\partial t} = 9 \frac{\partial^2 u}{\partial x^2}$ is [2]

- i) $u(x,t) = (C_4 \cos mx + C_5 \sin mx) e^{-9m^2 t}$
ii) $u(x,t) = (C_1 \cos mx + C_2 \sin mx) e^{-3m^2 t}$
iii) $u(x,t) = (C_1 e^{mx} + C_2 e^{-mx}) (C_3 \cos mx + C_4 \sin mx)$
iv) $u(x,t) = (C_1 \cos mx + C_2 \sin mx) (C_3 \cos ct + C_4 \sin ct)$

Q2) a) The first three moments $\mu_1^1, \mu_2^1, \mu_3^1$ about the value 2 of a distribution are 2, 32 and -80 . Find the mean, standard deviation and the coefficient of skewness of the distribution. [5]

b) A set of 10 coins are tossed 4096 times. Assuming that the coins are identical and fair, in how many cases do you expect: [5]

- i) 8 heads and 2 tails?
ii) at least 8 heads?

- c) Fit a Poisson distribution to the following data and find the χ^2 value.

x	0	1	2	3	4
f	122	60	15	2	1

Here f denotes frequency. Take $e^{\frac{-1}{2}} = \frac{3}{5}$. Round off frequencies to the immediate higher integer values. [5]

OR

- Q3)** a) 10% rivets produced by a machine are defective. Find the probability that out of 10 rivets chosen at random.

- none will be defective.
- one will be defective
- at least one will be defective.

Apply the Poisson random variable theory to solve this question. [5]

- b) In every 30 days rain falls on 10 days on an average. Obtain the probability that [5]

- rain will fall on at least 3 days of a week.
- the first 3 days of a week will be dry and the remaining 4 days wet.

- c) The monthly wages of 10,000 workers in a factory follows normal distribution with mean and standard deviation as ₹17000 and ₹1000 respectively. Find the expected number of workers whose monthly wages are between ₹16000 and ₹20,000. Take area $(0 < z < 1) = 0.34$ and area $(0 < z < 3) = 0.49$ where z is the standard normal variate. [5]

- Q4)** a) Prove the following identities (any one) [5]

i) $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$

ii) $\nabla^4 (r^2 \log r) = \frac{6}{r^2}$

- b) Find Directional derivative of $\phi = xy^2 + yz^2$ at $(2, -1, 1)$ along the line $2(x-2) = y+1 = z-1$. [5]

- c) Show that $\vec{F} = (2xz^3 + 6y)\hat{i} + (6x - 2yz)\hat{j} + (3x^2z^2 - y^2)\hat{k}$ is irrotational.

Find ϕ such that $\vec{F} = \nabla\phi$. [5]

OR

- Q5) a) For a solenoidal vector field $\vec{F}$, show that $\text{curl curl curl curl } \vec{F} = \nabla^4 \vec{F}$ [5]

- b) Find the directional derivative of $\phi = x^2 - y^2 - 2z^2$ at the point P(2, -1, 3) in the direction PQ where Q is (5, 6, 4). [5]

- c) Show that $\vec{F} = (y^2 \cos x + z^2)\hat{i} + (2y \sin x)\hat{j} + 2xz\hat{k}$ is irrotational Find ϕ such that $\vec{F} = \nabla\phi$. [5]

- Q6) a) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = x^2\vec{i} + xy\vec{j}$ and C is straight line $y = x$ joining points (0, 0) and (1, 1). [5]

- b) By using Gauss divergence theorem evaluate $\iiint_S \vec{F} \cdot d\vec{s}$ for vector field $\vec{F} = yz\vec{i} + zx\vec{j} + xy\vec{k}$ where S is curved surface of cone $x^2 + y^2 = z^2$, $z = 4$. [5]

- c) Use Stoke's theorem to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = e^x\vec{i} + 2y\vec{j} - \vec{k}$ where 'C' is curve $x^2 + y^2 = 4$, $z = 2$. [5]

OR

- Q7) a) Find work done in moving a particle around ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ in plane $z = 0$ where $\vec{F} = (3x - 2y)\vec{i} + (2x + 8y)\vec{j} + y^2\vec{k}$. [5]

- b) Use Gauss divergence theorem to show that $\iiint_V \frac{\vec{r}}{r^2} \cdot d\vec{s} = \iiint_V \frac{dv}{r^2}$ [5]

- c) Evaluate $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$ where $\vec{F} = (x^2 + y - 4)\vec{i} + 3xy\vec{j} + (2xz + z^2)\vec{k}$ over the surface of hemisphere $x^2 + y^2 + z^2 = 16$ above XOY plane. [5]

Q8) a) A taut string of length $2l$ is fastened at both ends, the midpoint of the string is taken to the height 'b' and the released from the rest in this position.

Obtain the displacement $y(x, t)$ if $\frac{\partial^2 y}{\partial t^2} = C^2 \frac{\partial^2 y}{\partial x^2}$. [8]

- b) Solve the one dimensional heat equation, $\frac{\partial u}{\partial t} = K \frac{\partial^2 u}{\partial x^2}$, subject to conditions. [7]

i) u is finite $\forall t$

ii) $u(0, t) = 0$

iii) $u(100, t) = 0$,

iv) $u(x, 0) = \begin{cases} x & \text{if } 0 \leq x \leq 50 \\ 100 - x & \text{if } 50 \leq x \leq 100 \end{cases}$

OR

Q9) a) If a string of length 4cm is initially at rest in its equilibrium position is set to vibration by giving each point a velocity,

$$\left. \frac{\partial y}{\partial t} \right|_{t=0} = \begin{cases} 3x & \text{if } 0 \leq x \leq 2 \\ 3(4-x) & \text{if } 2 \leq x \leq 4 \end{cases}$$

find the displacement $y(x, t)$

[8]

- b) An infinitely long uniform metal plate is enclosed between two parallel edges $x = 0$ and $x = \pi$ For $y > 0$. The temperature is zero along the edges $x = 0$, $x = \pi$ and at infinity. If edge $y = 0$ is kept at a constant temperature V_0 . Find the temperature distribution. $V(x, y)$ [7]

